224

EXPERIENCES ON THE LOW PRESSURE WATER HYDRAULIC SYSTEMS

Jussi AALTONEN, Kari KOSKINEN, Matti VILENIUS and Pekka KUNTTU
Institute of Hydraulics and Automation, Tampere University of Technology
P.O.Box 589, 33101 Tampere, Finland
(E-mail: jussi.aaltonen@cc.tut.fi)
(WWW: http://www.tut.fi/~ihatut/)

ABSTRACT

High and medium pressure water hydraulics is widely used in various industrial applications. In applications where extreme environmental safety is needed and in applications in fire hazardous environment water hydraulics has been proven to be the best solution available. However, component costs in high and medium pressure water hydraulics are relatively high and therefore they are often substituted by other methods of power transmission. Other alternatives, such as conventional oil hydraulics, pneumatics and electric drives, however always lack some qualities water hydraulics could offer. To offer a technically competitive and cost effective alternative for them, the possibility of using low pressure and relatively cheap materials and components, such as centrifugal pumps, in water hydraulic system was studied.

Test systems were constructed using centrifugal pump, pneumatic cylinders and ordinary plumbing components, including rigid PVC-tubing and PVC-hoses. The over all performance of systems was studied as well as performance and suitability of components.

Test results indicate that in applications where force required from actuators are not very high a low pressure water hydraulic system can offer a competitive alternative to conventional methods of power transmission. In this paper we present comparison of low-pressure water hydraulics to other alternatives of power transmission. Power density, system efficiency, relative component and system costs and noise level are compared, based on measurements from test systems. Also pressure transients and other characteristics of systems were studied and are presented in this paper.

KEY WORDS

Hydraulics, Water, Low-pressure, Centrifugal pump, PVC-tubing

INTRODUCTION

Conventional oil hydraulics and pneumatics, as well as electric drives are widely used in industrial applications. They are technically very advanced, reliable and generally cost effective ways of power transmission. They all have some serious disadvantages, such as environmental safety, power density, fire hazard and total efficiency. High and medium pressure water hydraulics can offer an environmentally and fire safe alternative for them, with high efficiency and power density [1]. However, they usually are not cost effective alternatives. In

Fluid Power. Forth JHPS International Symposium ©1999 JHPS. ISBN4-931070-04-3

conventional water hydraulics high-pressure forces to use relatively expensive materials, such as stainless steel. High pressure also leads to very small clearances in components, which means fine machining tolerances. Lowering the pressure makes it possible to use cheaper materials and rougher tolerances, which makes components significantly cheaper. Also making use of existing designs is possible. Low-pressure water hydraulics (LPWH) is a technology in between traditional oil hydraulics and pneumatics, in the sense of price, power density and efficiency. In the sense of environmental and fire hazard it is ultimate in comparison to oil hydraulics and electric drives.
Low-pressure water hydraulics is in nutshell [2]:

- Cost ≈ pneumatics
- Control ≈ hydraulics
- Force ≈ 3x pneumatics
- Environment ≈ safe

Total system costs will probably be more than in pneumatic system because of the need of return lines and separate power unit; pneumatic system usually utilises existing compressor unit. Total efficiency of the pneumatic linear actuator is roughly 10%, so if LPWH linear actuator could reach the total efficiency of 50%, the efficiency would quickly pay back the bigger initial cost. Furthermore the price of compressor unit is usually not taken into account when evaluating the cost effectiveness of a single pneumatic system.

OVERVIEW TO THE LOW-PRESSURE WATER HYDRAULICS

In materials handling oil hydraulics is many times used only because forces required are slightly more than pneumatics could offer, which leads to oversized systems. There is also always a risk of contaminating the product when using oil hydraulics. In process industry, such as papermaking and food industry it is extremely important that risk of contaminating the product is minimal, which makes water the natural selection for pressure medium. Also in any application in fire hazardous environment the water is clearly better solution than oil or compressed air.
Pressure range in LPWH-systems is 10-70 bar, clearly bigger than in pneumatics, but smaller than in high-pressure oil and water hydraulics. There is a wide range of components readily available for applications working in that pressure range.
VALVES – Diagram, poppet and other valves for ordinary plumbing and process piping are suitable within their pressure range. It is also possible to convert pneumatic valve designs to operate with water.
PUMPS – High-speed and multi-stage centrifugal pumps are suitable and relatively cheap power source. Piston and vane pump designs could also offer an alternative, but there are no cost effective alternatives readily available.
MOTORS – There are some high speed piston motors available for water hydraulics, but they are expensive due to their design for high pressures. For low pressure there is only one motor on market, Danfoss vane motor (50 bar, 2 kW) [3]. It is also the only true low speed motor available.
PIPES – Stainless steel pipes and fittings are too expensive to be used in LPWH-systems. Rigid plastic tubing and plastic hoses are a possible solution.
CYLINDERS – Many pneumatic component manufacturers already have corrosion resistant designs available, they are easily modified for use in LPWH-systems.
SYSTEM – In general an LPWH-system is by design as any other hydraulic system. Also controlling the system is possible same ways as in hydraulics in general. Availability of components restricts some control functions such as servo controls, as is the case in high-pressure water hydraulics also. Relatively high pressure transients, microbial growth, mineral matter depositing, relatively high seal friction and operational temperature range are points of pay attention in system and component design as in any other water hydraulic system. [2]

TEST SYSTEMS

Two test systems were build to get preliminary research data from components and performance of LPWH-systems.
Pressure range for systems was selected to be 10-20 bar, because of the components used. Both test systems utilise the same power source, 14 stage centrifugal pump.

- Flow @ 10 bar = 68 l/min
  Flow @ 20 bar = 33 l/min
- Max. pressure 26 bar
- Efficiency @ 10 bar = 0.38
  Efficiency @ 20 bar = 0.39
- Optimal efficiency 0.44 @ 16 bar, 50 l/min

Solenoid valves used were 2/2 poppet valves, designed for plumbing.

- Max. pressure 12 bar
- Body material brass
- Seal material NBR
- Internal parts AISI316
- Closing time 30 ms
- Opening time 30 ms

The tank of the system was completely sealed from operational environment to prevent air mixing to water and restrict microbial growth. Total volume of the tank was 100 l; 50 l of water in air pressure of 1 bar (abs) behind the diagram. The tank was standard plumbing component.

Two types of cylinders were used in test system 1. Both cylinders were designed for pneumatic systems. In test system 2 only cylinder 2 was used.

Cylinder 1:

- Max. pressure 8 bar
- Bore 32 mm
- Rod 12 mm
- Stroke 50 mm
- Body material polyamide
- Seal material NBR, polyurethane
- Rod material AISI 316

Cylinder 2:

- Max pressure 25 bar
- Bore 40 mm
- Rod 20 mm
- Stroke 400 mm
- Body material brass, anodised aluminium
- Seal material NBR
- Rod material AISI 316

Pipelines were constructed from rigid PVC tubing (rated 25 bar @ 20°C) in pressure lines or soft nylon reinforced PVC-hose (rated 10 bar @ 20°C) in return lines. Soft PVC is generally not acceptable material in water hydraulic systems [4]. Joints were made by gluing or by using brass compression fittings. In hose joints simple clamps were used. Ball valves in pressure lines were ordinary plumbing valves made from brass with AISI 316 balls. In return line ball valves were all PVC construction. Return line filter housing was of polypropylene construction. Filter element was made from continuous polypropylene fibre, absolute filtration rating 10 µm.

Pressure in both systems was controlled by by-pass parallel to pump; adjusting ball valve in by-pass line sets the maximum pressure and flow of the system. Both systems were controlled by simple programmable logic controller, so that systems could perform their work cycles continuously.

Function of the test system 1 (Fig 1) was to give data from the reliability of standard pneumatic cylinders in the water hydraulics. Both cylinders were run 10 000 strokes after which they were examined. Cylinders are used single action to give a simple way to monitor the leakage over the piston. Leakage in cylinders was monitored continuously. Also system dynamics was examined by measuring pressures and flow.

Test system 2 (Fig 2) was constructed to measure system dynamics in simple one axle linear movement; cylinder 2 was used in this system. Weights used were 0 – 70 kg, excluding weight of the linear table. This system differs from the system 1 by it having 2 metres of standard R1AT hose in between valves and power pack.

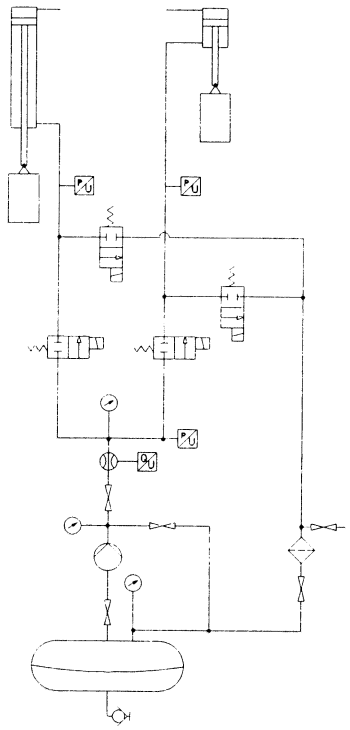


Figure 1 Test system 1.

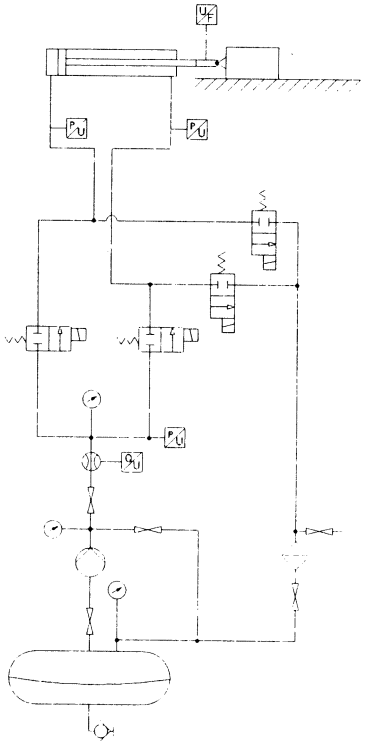


Figure 2 Test system 2.

In the test system 1 weights loading cylinders were simply hanged to the cylinder rod by flexible joint, no extra damping or forces, besides the cylinder friction were provided by mounting.
Load in the system 2 was placed on the linear table gliding on guides. Friction and viscous friction of the linear table causes damping relative to the load.
Data acquisition system in both tests consisted of PC-computer with 16-channel data acquisition card. Pressure and flow transducers were connected to the data acquisition card. In both systems flow transducer and supply pressure transducer were placed in the supply line, immediately after the pump. Load pressure transducers were placed cylinder lines. (Fig 1 and 2) Sampling frequency used in all measurements was 1000 Hz, due to limitations of transducers.

EXPERIMENTS WITH TEST SYSTEMS

In test system 1 cylinders were run 10 000 full strokes, in sequences of about 500 strokes. Test conditions were:

- Average piston speed
  Cylinder 1 60 mm/s
  Cylinder 2 120 mm/s
- Supply pressure 10 bar
- Weights
  Cylinder 1 60 kg
  Cylinder 2 60 kg
- Operating temperature 20 – 40 °C

During test period no leakage were observed in cylinder 2. Cylinder 1 leaked from beginning and towards the end of the test leakage increased, but never exceeded 5 ml / stroke. After the test cylinders were dismantled to visually observe wear. Cylinder 2 had no visual signs of wear; neither it had signs of corrosion. Cylinder 1 had some minor scratch marks in cylinder tube, not deep enough to cause the leakage. A manufacturing flaw in the piston caused leakage in the cylinder 1. Piston is moulded in single piece polyurethane plastic. Plastic injection channel mark was not removed carefully enough and caused seals of the piston to work improperly. No signs of erosion or cavitation wear were found in the components. In system components the only other drawbacks observed were two breakdowns in pipelines and pressure line valve of the cylinder 1 closing improperly.
Cylinder 2 was dismantled prior to test and all lubricant was removed from wetted parts of it. Cylinder 1 could not be dismantled without breaking; therefor it was not cleaned. As could be expected dismantling the cylinder 1 revealed heavy coat of lubricant in inner parts of cylinder, in full-scale system this could cause serious problems. No lubricants, especially vaseline based, are allowed in water hydraulic systems [4].

Pressure transients in the test system 1 were found to be relatively high, approximately half of the pressure level of the system. In figure 3 water hammer in the supply line caused by closing of the pressure line valve of the cylinder 1 can be seen. Fast closing valves always cause water hammers (pressure transients) relative to the length of pipe, closing time, speed of sound in the pressure medium and the specific weight of fluid [5]. Damping of pressure transient caused by water hammer effect is relative to mechanical characteristics of the pipe and acoustic characteristics of the system. Since test system does not have any elements causing additional impedance in the supply line the pressure transient is damped on frequency characteristic to the pipeline.

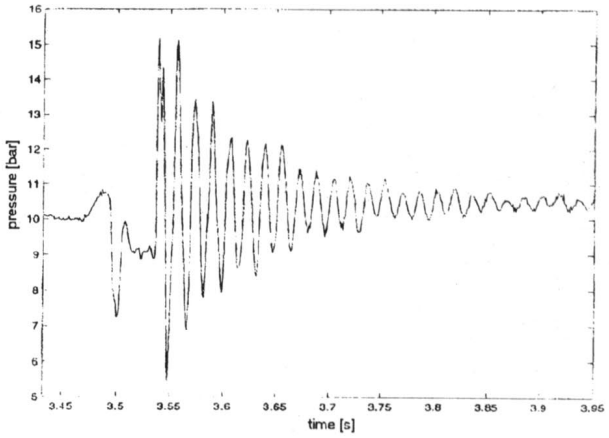


Figure 3 Pressure transient in the supply line (f_s = 1000 Hz).

Improper closing of the pressure line valve of the cylinder 1 can be seen in the figure 4. The uppermost curve represents the flow and the second curve is the pressure in the cylinder line. The third curve of the figure 4 is the supply pressure. When the cylinder reaches the end of its stroke the pressure rises quickly to the maximum for a brief moment before the return line valve opens. However, the pressure line valve closing causes a pressure transient to the supply line. Peak values of the transient are high enough to open the pressure line valve, thus causing a flow directly from the supply line to the return line. Dynamics of the flow sensor are unfortunately too slow for getting an accurate measurement of the flow during the pressure transients.

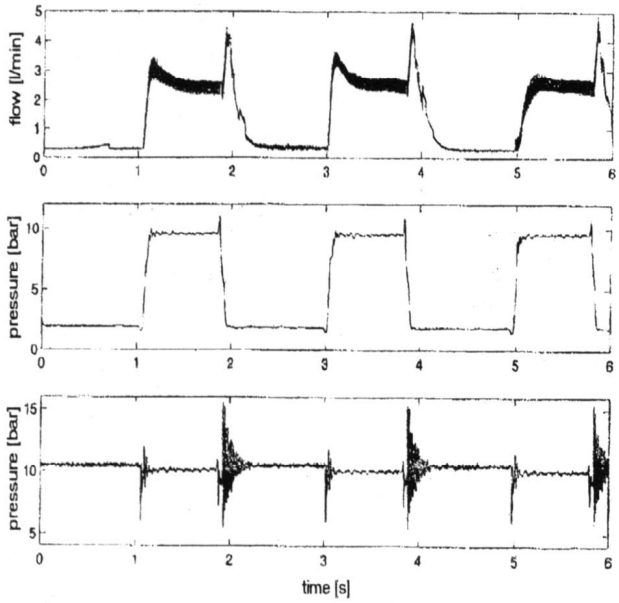


Figure 4 Pressure transients and abnormal function of the valve in the cylinder 1.

Tests performed with the test system 2 included some simple experiments with using 2/2-valves and cylinder 2 to drive a linear table. Test conditions were:

- Average piston speed
  70 mm/s @ 10 bar
  100 mm/s @ 15 bar
- Supply pressure 10 bar or 15 bar
- Load 0 – 70 kg + table
- Operating temperature 40 °C

The 2 metre hose in between valves and power pack lowers pressure transients in supply line, as can be expected. Pressure transients in cylinder lines and supply line when the cylinder reaches the rod side end without a load and with 10 bar supply pressure can be seen in the figure 5. Figure 6 represents the same situation as figure 5, but with the supply pressure of 15 bar and the load of 70 kg.

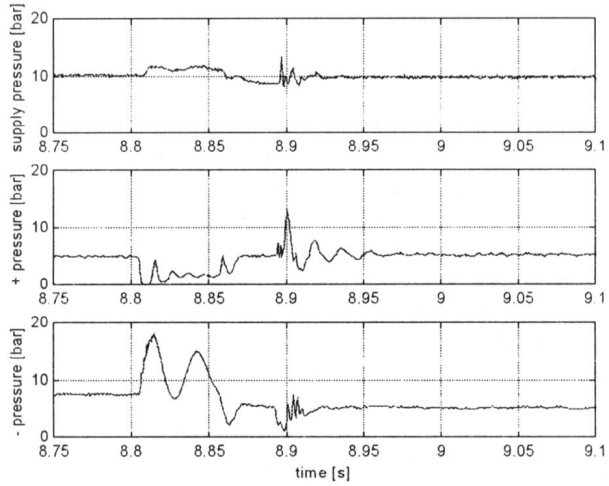


Figure 5. Pressure transients in lines as the cylinder reaches the rod side end without a load and 10 bar supply pressure.

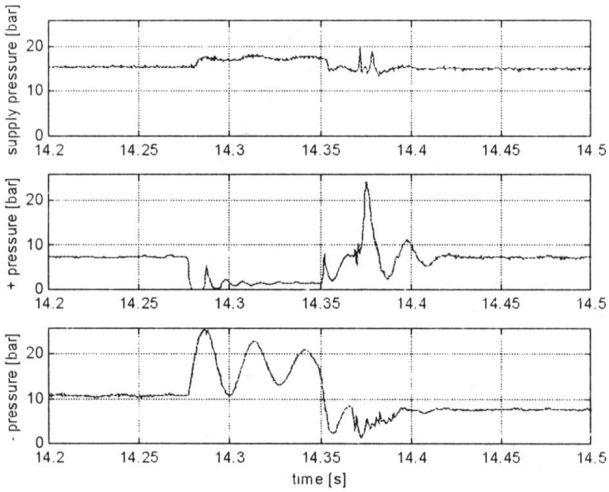


Figure 6. Pressure transients in lines as the cylinder reaches the rod side end with 70 kg load and 15 bar supply pressure.

Pressure transients in this test system are mostly caused by inertia of the system; thus they increase as the load is increased. From both figures can be seen a delay caused by logic controller, valves and sensors; as the piston meets the end it bounces back and sensor does not give long enough pulse to logic controller to switch valve function, the piston starts to fluctuate and as the fluctuation becomes slower the long enough pulse is got. Pressure transients caused by fluctuation are the biggest in cylinder rod side, pressurised side, but they are visible in supply line and piston side as well, but tank and tank line hose and pressure line hose damp them heavily. As the valve function switches closing the pressure line valve and opening the return line

valve on rod side and vice versa in the piston side valves and continuing small fluctuation cause pressure transients in all lines. Transients are quickly damped away.
In the test system all components used worked according to expectations. Some 50 – 100 strokes per single load test, 8 load tests in total, were run.
In both test systems getting air out from system was found to be a problem. Same methods than in oil systems do not apply because air is unsolved from water much more rapidly than from oil. Also small air bubbles join more quickly in water forming larger air pockets.

CONCLUSIONS

Two test systems build give a good example of problems in the low-pressure water hydraulics. Pressure transients are generally 30% higher in water hydraulic system than in oil hydraulic system [5]. In the case of LPWH-systems the pressure transients may be 50% - 100% of the pressure level of the system, which means they have to be taken into consideration during system design. In these test systems rigid PVC-tubing was used. Rated pressure for tubing is 25 bar @ 20°C. However, system temperature was during tests much higher, which makes tube rating to decrease. In real scale system this tube could not have been safely used without lowering transients significantly. In tests it was also found that glue joints could not handle the temperature and pressure, but two breakdowns occurred in joints. After breakdowns the new pipelines were supported better, which helped, but can not be considered as a solution for breakdowns in joints. One way to damp pressure transients is to use long hoses as in the test system 2.
Standard brass compression ring fittings gave a reliable and leak free joints with PVC-tubing. In general brass and stainless steel plumbing components are cost effective to be used in LPWH-systems. Their pressure ratings usually are 10 or 16 bar, but as practice has shown they safely hold much greater pressures. Of course for them to be utilised in full scale systems further testing must be done.
Pressurised tank is an effective way to prevent air mixing to water and control microbial contamination. But in systems having pressure level this low, the return line pressure increases too much, thus decreasing efficiency. Using lower pressure in the tank and larger tank would remove this problem but make tank too large.
Using standard plumbing poppet valves of compact outer dimensions is not possible. Most of the poppet and diaphragm valves have too small K_v-value, as did valves used in tests. To get system efficient the pressure drop in valve should only be some 1-2% of the pressure level of the system. It was also found that valves suffer reliability problems when operated near maximum pressure with relatively high frequency.
Multi-stage centrifugal pump works well as a power source for LPWH-system. Its characteristic curve is well suited to hydraulics on its flat high-pressure end, giving only small changes to flow as the pressure changes. One advantage is no need for pressure relief valve, but only a simple by-pass to control flow and pressure. Significant disadvantage is efficiency and lack of relatively small pumps. If pump is working outside of its optimal pressure range and flow must be limited with by-pass the total efficiency of power pack will be low.
Pneumatic cylinder is, with few modifications, a suitable linear actuator for LPWH-systems. However, materials usually used in pneumatics are not always suitable, even though they mostly are corrosion resistant. For example in itself anodisation gives a very good corrosion protection for aluminium, but it is very fragile and gives minimal coat on edges and holes, thus damaging it and revealing unprotected aluminium in installation is very likely. Also using materials of very different electro-negativity value, like aluminium and brass, is not a good practise. Polyamide used in cylinder 1 is also slightly suspicious material choice for water hydraulics, due to its tendency to absorb water and swell. Reliability of cylinders can not be determined due to shortness of tests, but during tests wear was minimal and therefor cylinders seem very promising.
Comparing the test systems to the systems made to perform similar tasks with other technologies makes LPWH-technology to look promising. The powerpack used could offer power to 12 pieces of either system 1 or system 2 with efficiency of 38%. Using better sized valves and normal tank the efficiency of pneumatic linear drive could easily be doubled and also some other advantages such as greater stiffness and better fire safety can be got; with these components only slightly bigger forces could be achieved. Price compared to pneumatic system is more or less the same, if the power source is taken into account in both systems. In comparison to other technologies the price is clearly cheaper and advantages greater.
Noise level of the whole system was in every test clearly under 80 dB. Only one oil hydraulic gear pump on typical pressure range produces approximately 70-80 dB and water hydraulic pumps and compressors are even noisier. Only electric drives can achieve same noise levels without special arrangements.

SUMMARY

1. Pressure transients in LPWH-systems need special attention in component selection and system design.
2. Multi-stage centrifugal pump is suitable to be used as a power source in LPWH-systems.

3. Valve technology for LPWH still needs to be developed further.
4. With small modifications pneumatic cylinders can be used in LPWH-systems.
5. Rigid plastic tubing is a suitable material for pipelines, but joining techniques need development.
6. Most of standard plumbing components are well suited in LPWH-systems working within or slightly above their pressure range.
7. The noise level of LPWH-system is relatively low.
8. Pressurised tank is not suitable for low-pressure systems.

ACKNOWLEDGEMENT

The assistance of Mr Max Lakkonen in the measurements of the test system 1 is greatly acknowledged.

REFERENCES

1. Koskinen K.T., Vilenius M.J., Water as a Pressure Medium in Fluid Power Systems, IFAC Workshop on Trends in Hydraulic and Pneumatic Components and Systems, Chicago Illinois, 8-9 November 1994, 13p.
2. Kunttu P., Low Pressure Water Hydraulics, Fluid Power Theme Days in IHA, Tampere Finland, 25-26 September 1997, 6p.
3. Jepsen H., Tap Water Hydraulic Vane Motor and Valves for Medium Pressure and Low Power Systems, International Workshop on Water Hydraulic Systems and Applications, Copenhagen Denmark, 3-4 September 1998, 3p.
4. Aaltonen J.M., Soini S.M., Koskinen K.T., Vilenius M.J., Water Quality Control in Fluid Power Systems Using Tap Water as a Pressure Medium, SAE Off-Highway and Powerplant Congress and Exposition, Milwaukee Wisconsin, 14-16 September 1998, 6p.
5. Trostmann E., Water Hydraulics Control Technology, Marcel Dekker Inc, New York, 1996, 180p.